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Data-Driven Fast Clustering of Second-Life Lithium-Ion Battery: Mechanism and Algorithm

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While electrical vehicles (EVs) are expanding rapidly and getting more and more popular in the market, researchers have started to leverage the remaining capacity of used or to-be-retired batteries for their second-life applications. It is crucial to develop a fast and efficient technology to first sort them and then extend their life while delivering energy, waste reduction, and economic benefits. In this work, a pulse clustering model embedded with improved bisecting K-means algorithm is developed to effectively sort retired batteries with life cycles ranging from new to an end-of-life state. The relevance of selected variables is rigorously validated, reaching the accuracy as high as 88% compared with the traditional full charge–discharge test. To note, the test time has largely reduced from hours to minutes. This data-driven clustering modeling with fast pulse test is a promising approach for clustering lithium-ion batteries, which is demonstrated with a home-built and high throughput intelligent clustering machine. In general, the technology opens a new generation of battery clustering, improving the efficiency and accuracy over the past semiempirical approaches.

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1. Introduction

Lithium-ion batteries (LIBs) have a very wide range of applications due to their low price, decreasing cost, high energy density, and long lifetime.[1,2] However, owing to the current manufacturing technology, batteries currently used in electric vehicles (EVs) do not maximize the life of LIBs.[3,4] Generally, a large proportion of batteries retired from EVs have about 80% capacity of their initial capacity.[5,6] Accompanying with aging, LIBs experience voltage decay, resistance increase and capacity loss, causing the obvious difference among retired batteries.[7] Taking battery thermal management as an example, different degrees of battery aging will increase the difficulty of thermal management and increase the risk of battery use.[8–10] Therefore, accurately clustering the usable batteries with the similar performance will open a new door for second-life of these retired LIBs.[11] Similarly, reused batteries will undergo a clustering process, but due to different working conditions where the loaded current, temperature[12] and state-of-charge[13] change dramatically. The impact on the degradation and the remaining life of the battery is different[14–16] and it is in great need to develop fast clustering technology.[17] At present, approaches using data-driven[5,18] and machine-learning techniques enhanced high-throughput method to predict battery life has been demonstrated with reasonable results.[19] However, they need to use the full charge–discharge method, which is extremely time consuming.[20–22] In contrast, external short pulse test has been employed to estimate the internal electrochemical state such as internal battery potentials, concentration gradients and state-of-health (SOH).[23] The rapid clustering, life prediction, and consistency selection of retired lithium-ion batteries are very challenging due to no historical data record[24] and nonlinear lifetime decay of batteries.[25,26] Therefore, variables are different, even under the same work environment, leading to cluster LIBs of the same performance is extremely difficult, especially for retired batteries.

Many previous studies have revealed the aging mechanism of lithium batteries,[21,27–29] mainly caused by the solid-electrolyte inter phase,[30,31] the loss of active material,[21] the generation of dead lithium[32] and the increase of internal resistance.[33,34] Currently, the clustering and the prediction of remaining useful life (RUL) are mainly semiempirical methods based on aging mechanisms above.[35,36] More accurate tests such as electrochemical impedance spectroscopy[37–39] and coulomb efficiency[40] can reflect battery aging in certain situations. However, from the perspective of efficiency and cost, these methods are obviously not suitable for the fast and efficient clustering.[41] Therefore, it is urgent to develop an efficient clustering method to increase the efficiency for LIBs' second-life reusing.

The data generation has raised statistical techniques to study the mechanism and even to predict the cycle life of lithium ion batteries.[5,42–44] This new approach has accelerated progress in different area, including material discovery for energy storage,[45] identification of synthesis methods,[19] prediction of material properties[46] and discovery of new catalysis.[42] Progresses have been reported on applying data-driven techniques to predict state of health with data collected in both laboratory and industry,[43,47] though these data are obtained by the conventional method via testing several cycles or even hundreds of cycles to ensure the accuracy without abnormal samples.[18]

Herein, a pulse clustering model embedded with improved bisecting K-means algorithm has been developed to effectively sort out retired batteries with life cycles ranging from new to the end-of-life states. At the same time, the proposed clustering method is integrated into high throughput intelligent sorting machine. The 130 cells with cycle lives ranging from 50 to 800 has been generated (the cycle life is defined as number of cycles until 50% of nominal capacity). The pulse data has been compared with the traditional charge–discharge test,[48,49] reaching the accuracy of up to 88%. Especially, the sorting time has been greatly reduced from hours to 2 min, greatly improving the efficiency of battery clustering. Moreover, when the clustering rules are selected by removing the batteries that are difficult to separate to ensure consistent performance of the same type of battery, higher accuracy can be achieved which is applicable in industry. Based on fast pulses, intelligent sorting machine has been designed with improved bisecting K-means algorithm. To note, this method does not require battery history usage data and calibration of the label, which is quite different from other existing battery performance prediction and typical classification approaches.

1.1. Data Generation and Key Parameters Acquisition

The data contains a full range of variables for lithium batteries is expected, ranging from brand new to retired batteries. To probe the data, 18 650 cylindrical batteries (NMC/graphite) (Matsushita) were cycled in a controlled temperature environment at 25 °C under varied charging conditions (see Experimental Section for details). By deliberately varying the battery condition, a data-set is generated that captures a wide range of cycles lives, from new to 800 cycles (average cycle life of 750 cycles). While the test temperature is controlled in 25 °C incubator, the effect of environmental temperature on battery performance is negligible.

Figure 1a shows the discharge capacity as a function of cycle for all the tested batteries, where different colors represent varied cycle life. The color of each curve is scaled by the battery's cycle life. After each 50 or 100 cycles, the current has been reduced to obtain capacity at 0.5 C rate (C-rate is the measurement of the charge and discharge current with respect to its nominal capacity: C-rate = current/rated capacity). It can be seen that the battery retention ratio and consistency are better when the battery is cycled in the starting cycles, and the difference between each cell become larger with the increasing cycles.[50,51] Figure 1b shows the pulse voltage distribution of 136 cells under different capacities. As can be seen from the pulse data, a better the capacity has more concentrated voltage. Similarly, the difference between each battery increased after aging. Figure S1 (Supporting Information) shows the capacity distribution between 550 and 800 cycles, the battery aging speed is not the same, which results from varied battery degradation performance. In Figure S2 (Supporting Information), 10 batteries are token as a sample test. The initial consistency is good and the 0.5 C capacity difference is only 1.3%. After accelerating aging test (cycling at 2 C rate), when the cycle is 500 cycles, the remaining capacity is 80%. The inconsistency increases, the capacity difference is greater than 6%, and it needs to be reorganized after clustering. While, the samples are roughly divided into two groups, the error range is reduced (Table S1, Supporting Information). The difference of battery capacity becomes larger with increasing number of cycles.[52] When these similar batteries are clustered into one category, the difference becomes smaller. Hence, obviously, a reasonable clustering can improve the consistency of the battery. Figure 1c shows a detailed cell charge–discharge curve, the decreasing speed of capacity enlarges obviously. On the other hand, in Figure 1d, for the same cell, the change of the pulse curve (every hundred cycle) is consistent with capacity decay. Comparing Figure 1c with d, as the battery cycle increases, the change in battery pulse voltage is somehow consistent with the attenuation of battery capacity. Through the study of the stabilization curve, the same trend is found. Our former work has proved that the voltage stabilization curve reflects the state of health of lithium-ion battery.[7]

Given this simple trend, a data-driven approach is employed considering the relation between the full charge–discharge data and the pulse data.[23] From the pulse test, the resistance and voltage stabilization information could be observed, meanwhile, the full charge-discharge test in traditional sorting approach is used for validating the accuracy of pulse method. As shown in Figure S3 (Supporting Information), the pulse test is much faster than the traditional full charge and discharge test. It only takes less than 120 s, while the traditional method lasts for hours.

Note that, LIBs are subjected to three states: charge, discharge and rest. The voltage stabilization[53] in the rest stage are identified and used to assess the SOH of LIBs, the method is elaborated in Figure S4 (Supporting Information). When the battery is rest, the voltage will partially return to a stable state. This curve of different state batteries will behave differently, which is called the voltage relaxation behaviors.[53] From the Formula (1), this curve could be fit to get the corresponding parameters:y_0 (a stabilization curve coefficient related to open-current voltage); A_1 (a stabilization curve coefficient); ${}^{t}{}_{1}$ (Stabilization curve coefficient) and R_1 (Resistance calculated by pulse curve).[7]

$$y = y_0 + A_1 e^{-x/t}{}_1 \tag{1}$$

So far, capacity (Q), voltage (U) and resistance (R) are mostly used parameters describing the SOH of LIBs. Researchers have reported several works on capacity evaluation, voltage change, and resistance identification and their evolutionary performance

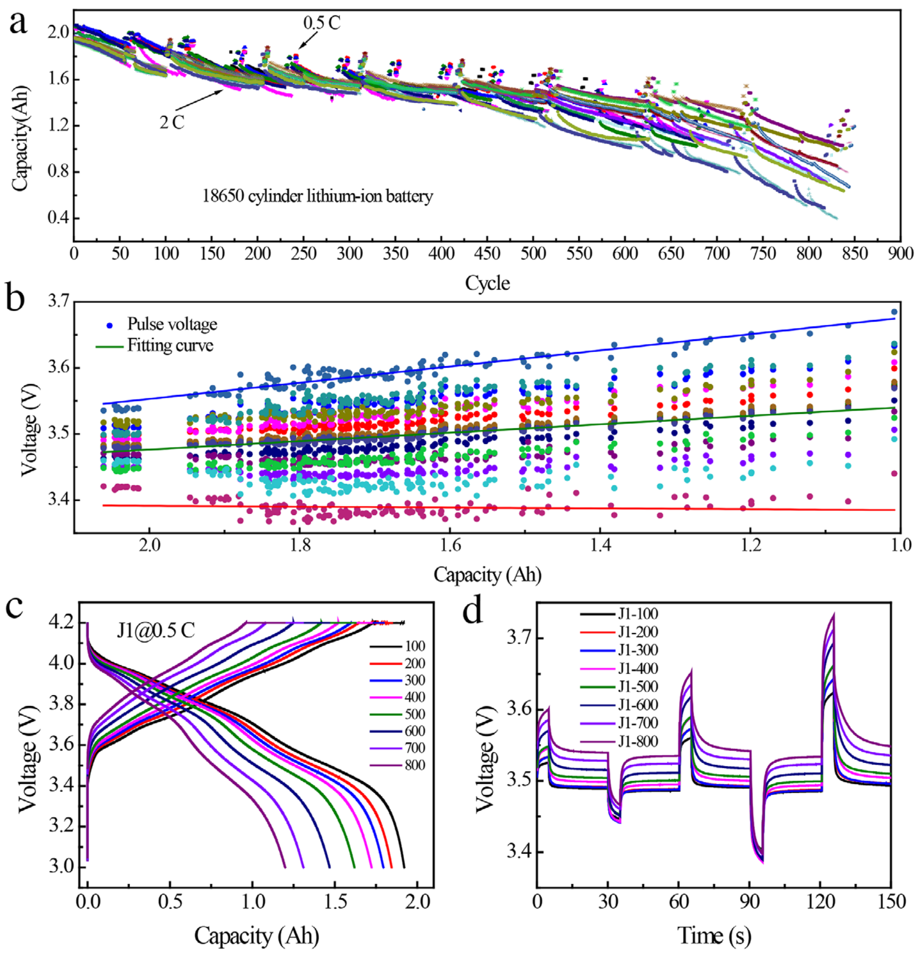

Figure 1. The electrochemical performance of total features designed for clustering test. a) Discharge capacity for 800 cycles of NCM/graphite cells. The color of each curve is scaled by the battery's cycle life, as is obeyed the manuscript. After every 50 or 100 cycles, the current is reduced to obtain the actual capacity. b) The pulse voltage distribution under different capacities. c) J1 cell charge–discharge curve at 0.5 C rate, there will be a significant reduction in capacity per 100 cycles. d) The pulse curve collected per 100 cycles for the same cell J1, the upward trend of pulse curve is consistent with the change of capacity decay at each one hundred cycle. As can be seen from the pulse data, a better the capacity has more concentrated the voltage but the worse the capacity has more dispersed the voltage. Hence, voltage is an effective response to battery performance.

when cycling or storage in different types of LIBs.[49,50] The correspondence between the pulse acquisition features and the full charge–discharge obtained features shown in Table S1 (Supporting Information). According to the corresponding variables and the meaning of the variables, stabilization curve coefficient (y_0), first dimension after pulse dimension reduction (F_1), and resistance calculated by pulse curve (R_1) are selected as the clustering indicators, and selected the full-charge and discharge $U/R/Q$ as the verification index. Their corresponding correlation among the parameters, the clustering performance and mechanisms will be discussed in detail later.

1.2. Data-Driven Clustering Approach

Unsupervised clustering techniques have been applied to solve many important problems.[54,55] Especially, the K-means clustering algorithm is one of the most widely used algorithms in cluster analysis. However, it mainly faces two kinds of drawbacks: the determination of cluster number K and the determination of initial cluster centers.[56] If these initialization parameters are not handled properly, the algorithm will fall into local optima, becoming very sensitive to abnormal samples.[57] In reality, the exact number of clusters in a batch of retired batteries is always unknown. It is difficult to gain the knowledge about these initialization parameters from real-world experience. So other methods based on the data itself need to be introduced.

Here, a canopy algorithm is used to gain the cluster number. This algorithm is a preclustering algorithm introduced by Andrew McCallum, Kamal Nigam, and Lyle Ungar in 2000. It is often used as a preprocessing step for the K-means algorithm. The main process of the canopy algorithm is shown in Figure S10 (Supporting Information). To further deal with the problem of initial cluster centers, a bisecting K-means algorithm is applied, which largely increases the stability of clustering result. In each iteration of the bisecting K-means algorithm, the cluster center is chosen based on the current cluster density.[58] The main process of the bisecting K-means algorithm is shown in Figure S11

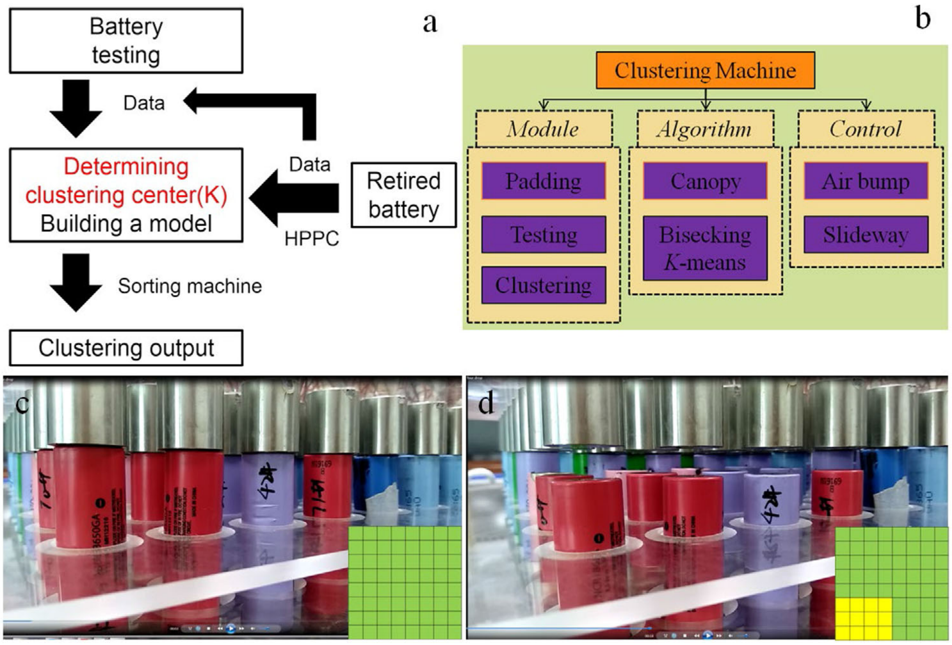

Figure 2. Implementation path of high-throughput intelligent sorting. a) Schematic illustrating a high throughput intelligent sorting machine system. b) Composition of clustering machine. c)The pristine state and d) six batteries are clustered.

(Supporting Information). Combining the above algorithms, an improved bisecting K-means algorithm based on canopy is implemented for retired battery clustering. The whole logic flow is illustrated in Figure S12 (Supporting Information).

To realize the aim of fast clustering, the implementation path of high-throughput intelligent sorting is designed. **Figure 2**a show the schematic of a high through output intelligent sorting machine system. It is a closed loop optimization system. First, the battery is tested, and the test data is used to establish an early clustering model, through the canopy algorithm to establish a reasonable cluster K, through the bisecting K-means algorithm to determine the cluster center. Repeat the program until the most reasonable clustering. For decommissioned batteries, HPPC tests were used to quickly obtain electrochemical data, input into the model, and quickly sorted. With this method, it is possible to eliminate the need for long loops and improve the clustering efficiency of retired batteries. At the same time, the new data can further optimize the model. The cooperation of machine learning clustering algorithm and intelligent sorting machine will solve the problem of reuse of retired batteries. The structural model of clustering machine is shown in Figure 2b, which includes padding module, pulse testing module and clustering module. Figure 2c,d shows the clustering module with the output of clustering results inserted. Figure 2c shows the pristine state and six batteries are clustered in Figure 2d. They are all cut out from the control video (Figure S14 Video, Supporting Information). The control part combined with the clustering algorithm can solve the ladder clustering problem. The human and material resources can be reduced because 64 or more cells can be tested at the same time.

In this work, the data from the pulse test is used for training the model while the data from the constant-current and constant-voltage (CC-CV) charging and constant-current (CC) discharging test is used for validation. Here, one pulse curve is regarded as one sample $X_i = (X_i^1, X_i^2, \ldots, X_i^n)^T$ where X_i^t represents the voltage value at time point t. In average, there are around sixty thousands of points in one pulse curve. In charge stages of the pulse test as shown in Figure S3 (Supporting Information), the voltage immediately reaches a high level and then slightly changes. While in the rest stages, the voltage values remain the same. In short, there exists a lot of redundant information in the pulse test data. Moreover, our former work investigated the voltage relaxation curves to evaluate the state-of-health (SOH) on lithium-ion battery.[7] So it is necessary to simplify and extract useful information from the original data of pulse test.

To reduce the dimension of each pulse sample, a principal component analysis (PCA) method is adopted. PCA is one of the most classical techniques to reduce the dimension of the data set. It maps the original samples onto principle component vectors so that the number of features is reduced to the number of principle components, while it still maintains most of the relations between samples. Figure S13 (Supporting Information) shows the process of PCA. In this paper, the largest eigenvalue from a PCA process covers over 95% of the variance of the whole data set. So a new feature F_1 is generated from the mapping of original data onto the eigenvector corresponding to the largest eigenvalue. Besides the feature F_1, there are two more features calculated from the pulse test data: y_0 from voltage stabilization curve and R_1 known as internal resistance. The detailed process of feature selection is available in Table S2 (Supporting Information). In conclusion, $F_{1/}y_{0/}R_1$ are used as new features for later clustering, while the validation is based on traditional $U/R/Q$ features.

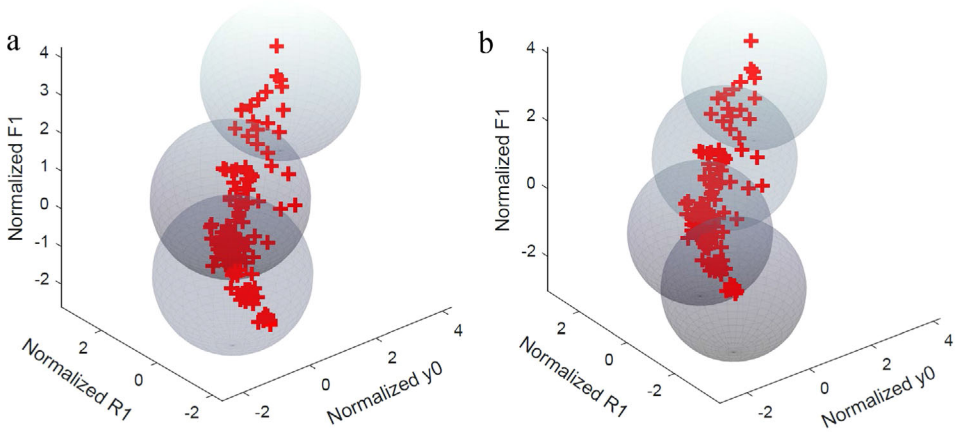

Figure 3. The clustering results of canopy algorithm. The features chosen to cluster are the F1feature from the PCA of pulse test, R_1 calculated from the pulse curve and y_0 came from voltage stabilization curve. The canopy algorithm is used to obtain the battery data clustering situation, which are a) three categories and b) four categories. On this basis, the bisecting K-means algorithm is used for clustering, and the relevant accuracy is checked.

1.3. Performance and Sorting Models

The reported remaining life time prediction or other techniques to sort or analyze the quality of battery usually require the life-long information and long-time testing, which is time consuming and expensive. The battery is clustered without requiring data about historical usage of battery. First, the battery is discharged and then charged to 5 % state-of-charge (SoC) to make all the tested battery have the same condition. In the meantime, the data is observed in 10% SoC (Figure S7, Supporting Information). However, the accuracy is reduced compared with 5% SoC. In order to study the reasons, Figure S8 (Supporting Information) shows the y_0 and internal resistance features of the sample at different SoC. The results of 10% and 15% SoC are consistent, slightly lower than 5% SoC. The accuracy changes from 88.97% to 87.50%. The reason is that when the SoC increases from 5% to 15%, the open current voltage (OCV) approaches the voltage platform, which means the lower SoC can reflect battery variability better. Thus, the pulse data at 5% SoC is selected to make sure all the data comparable. So, in later discussion, the results are all based on 5% SoC data-set.

Because of the random initialization of canopy algorithm, it often has several different results. However, it will largely narrow down the reasonable range of *K*. As for the data set in this paper, **Figure 3** shows the result of canopy algorithm. There are only two different conditions: $K = 3$ and $K = 4$. To further decide the exact value of *K*, Calinski-Harabaz (CH) index can be calculated. CH index for $K = 3$ is 204.86 while $K = 4$ is 269.63. Therefore, $K = 3$ is used for later clustering.

Then, the bisecting K-means is used and the accuracy combined with validation result is checked. **Figure 4**a shows, the comparison of the result is based on new features $F_1/y_0/R_1$ and on $U/R/Q$ when $K = 3$. The inner core (*C*) represents the result based on new features, while the outer shell (*C*-real) represents the result from $U/R/Q$. If a sample has same color between its core and shell, it is correctly clustered, vice versa. So, when $K = 3$, the accuracy reaches 88.97 %, while $K = 4$, the accuracy is 82.35%. (See detail in Table 1) The accuracy of the every pair indicators of verification can be obtained by using $R - U$ (Figure 4b), $Q - U$(Figure 4c) and $Q - R$ (Figure 4d) as verification indicators respectively. In the three figures, the verification of different dimensions has obtained a good accuracy. The verification results of $K = 4$ is shown in Figure S6 (Supporting Information).

Table 1. Clustering results shown in Figure 3.

Battery clustering	Clustering accuracy		
	Number of categories	Number of samples	Accuracy
	3	136	88.97
	3	136	82.35

The samples refer to data used to cluster with pulse data and evaluate with full charge and discharge feature, respectively. The accuracy in the test is the data including the new and the bad patterns. Therefore, if the testing abnormal data is erased, the accuracy can be much higher.

To further understand this high accuracy, the mechanism identification of pulse test is established to understand the relationships between new features $F_1/y_0/R_1$ and traditional features $U/R/Q_{\text{in}}$ **Figure 5**. It shows nearly linear correlations $Q - F_1$ (Figure 5a), $y_0 - U$ (Figure 5bf) and $R - R_1$ (Figure 5c) and ensures the high accuracy. The *R*-square of $Q - F_1$ reaches more than 0.93 and F_1 feature from pulse test contains the same property of capacity from full charge-discharge test. On the other hand, y_0 and R_1contains the same information of voltage and resistance. Thus, the linear relationship is easy to understand and the R-square of both is higher than 0.82. However, the inner reason for $Q - F_1$ relationship lays on the aging process of batteries. Incremental capacity (IC) have been used for understanding degradation mechanisms.[59] In addition, the capacity loss in 18650-type NMC/graphite cells is proved due to the loss of cyclable lithium.[60] **Figure 6**a shows capacity incremental curve of a random cells. It can be clearly seen that the aging battery peak will gradually move up and cyclable lithium is reduced,[61,62] resulting the changing position of extreme points. The peak polarization of IC curves correlate well with the capacity retention data.[63] Similarly, in Figure 6b, the pulse feature F_1 has a nearly linear correlation with these extreme points in Figure 6a. Thus, F_1 can also re-

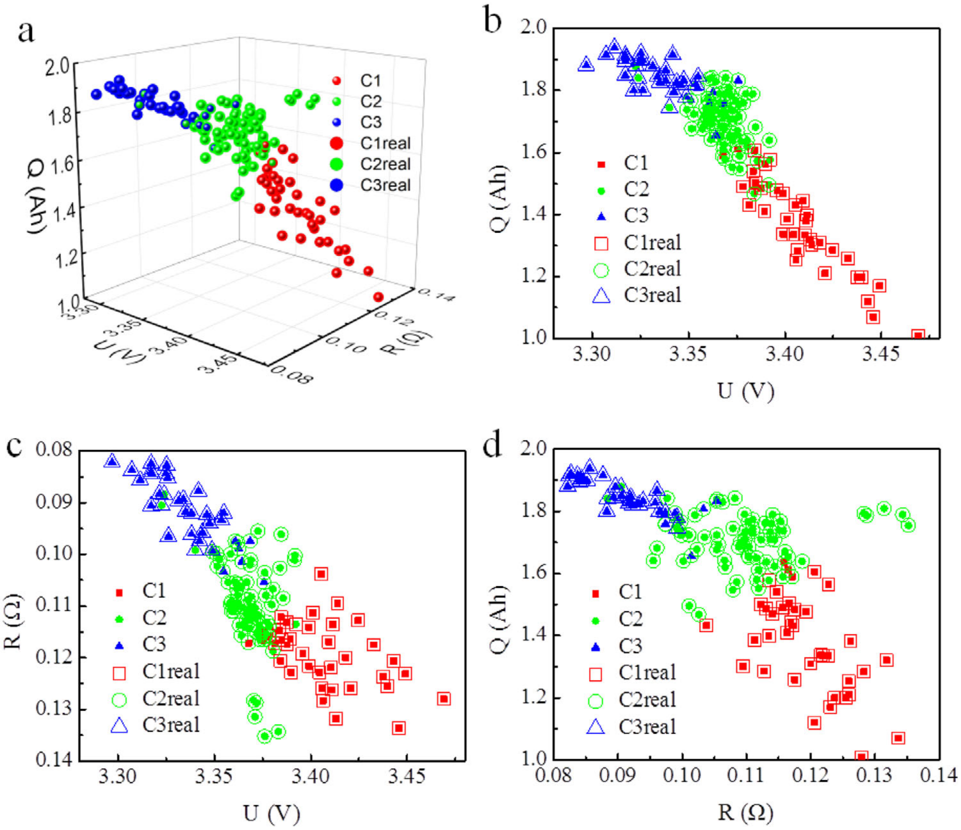

Figure 4. The clustering results. **a**) The dots are the clustering results with pulse data (C) and the electrochemical data obtained by full charge–discharge (C real), the same color represents the clustering is accurate. Two different color points indicate incorrect clustering. It can be seen that the clustering pulse test data classifies the battery, and the reliability exceeds 88% at three classes in this data-set. The accuracy of the two indicators of verification can be obtained by using b) *Q*–*U*, c) *R*–*U* and d) *Q*–*R* as verification indicators respectively. They all have high accuracy.

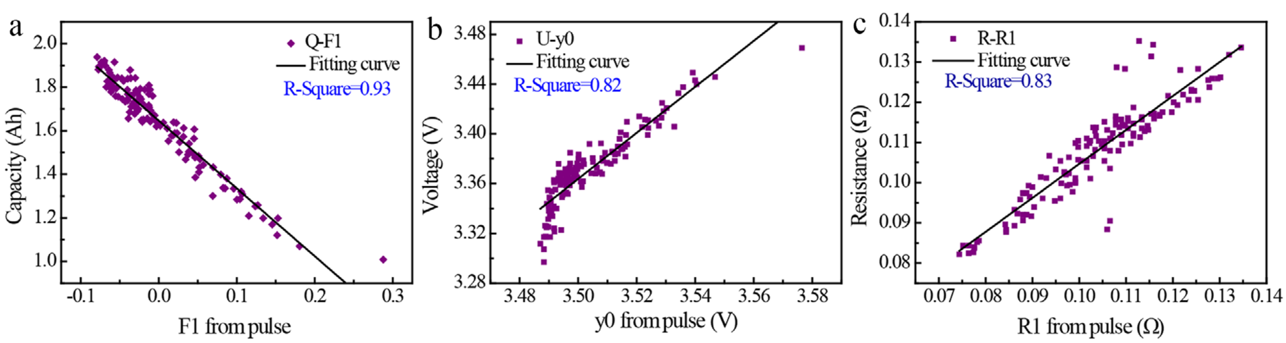

Figure 5. The mechanisms of pulse test clustering. The pulse test data are used to collect the feature of different stage of battery. And we use the pulse test data to go through the clustering algorithms and using the tradition charge and discharge data to verify the accuracy. The correlation between clustering indicators and verification indicators is as follows: **a**) Correlation between *Q* and F_1, b) *U* and y_0, c) resistance and R_1 are nearly linear (except there is a decaying form at Figure 5b when the y_0 are the lowest points when battery has extremely low capacity). The nearly linear relationship is the direct cause of achieving higher accuracy.

flect the capacity retention or degradation, directly related to the aging of batteries. Hence, the F_1 feature stand for same meaning of *Q* while the testing test is reduced a lot. In the meantime, we also observed other feature's relation. The results show that the correlation is not very good, (see detail in Figure S5, Supporting Information). Hence, the reasonable features $F_1/y_0/R_1$ from pulse test could be decided to cluster retired battery fast.

Back to Figure 4, it is clear that most error points locate in the boundary of two clusters while some error points result from abnormal test. To further understand why these error points exist, some characteristics of pulse tests are shown in Figure S8 (Supporting Information). In short, when the performance of two batteries is very similar, tests are not so distinguishable that the errors mainly locate in the boundary of clusters. However, in real-

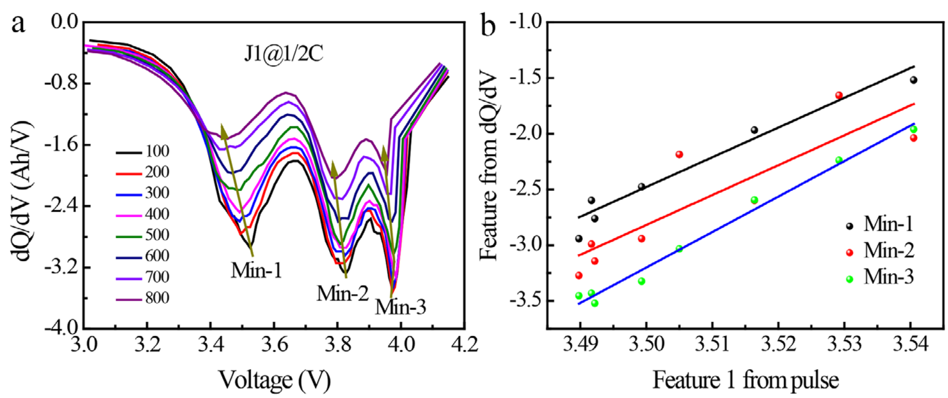

Figure 6. Mechanisms of pulse test. a) Capacity incremental curve, it can be clearly seen that the aging battery peak will gradually move up, and live lithium is reduced. The voltage change of the pulse curve and the stabilization curve is the same as the change of the capacity increment, so the pulse voltage curve has theoretical support as a fast clustering index. b) The pulse feature F1 has a nearly linear correlation with the incremental capacity.

world application, those samples located in cluster boundaries can be directly abandoned so that the final accuracy can be much higher.

2. Conclusions

In summary, a data-driven fast clustering approach of retired lithium-ion battery is developed. Compared with the traditional charge-discharge test, the time using pulse test is reduced from hours to minutes with the accuracy of 88%. The fundamental mechanisms are explored, in which pulse test feature is consistent with charge–discharge test for the aging of lithium-ion battery. Moreover, it is found that pulse feature reflects active lithium-ion in a battery system. In general, our approach of battery clustering improves the efficiency and accuracy of semiempirical methods. This work develops a time-saving clustering method and the application of data based high throughput intelligent sorting machine can promote the practical application of lithium battery.

3. Experimental Section

Cell Cycling and Data Generation: Thirteen commercials 18650 cylinder high-power NCM/graphite cells were used in this work. The cells have a nominal capacity of 2.1 Ah and nominal voltage of 3.6 V. The manufacturer's recommended fast-charging protocol is 2 C constant current-constant voltage (CC-CV).

All cells were tested in cylindrical fixtures on LAND battery testing cycler. The tests were performed at a constant temperature of 25 °C in a specific humidity.

In this work, cyclic test was at the current of 2 C and capacity was collected at 0.5 C. Moreover, for each battery, its open-circuit voltage (OCR) was approximated as the final voltage in the rest period, and its internal resistance (R) was derived through dividing the transient charging voltage by the current, and then, the three verification parameters $U/R/Q$ of each battery were obtained with this full charge-discharge test. In the meantime, the batteries were tested via pulse test. The detailed pulse steps are as follows:

1) Charge–discharge with a 0.5 C (0.5 A) current for 5 s, then rest for 25 s each.
2) Charge–discharge with a 1 C current for 5 s, then rest for 25 s.

Clustering Algorithm: Before proposing the improved bisecting K-means algorithm, there are two basic algorithms need to be introduced first: K-means and bisecting K-means algorithm.

The K-means algorithm is one of the most widely used partition clustering algorithms. It is easy to implement and is with a fast convergence speed. Based on the similarity between samples, the data-set is divided into K clusters, while the value of K needs to be determined. And the similarity is often defined as the Euclidean distance between data points. The distance and cost functions are shown below.

$$d(x_i, c_j) = \sqrt{\sum_{k=1}^{n} \left(x_i^k - c_j^k\right)^2} \tag{2}$$

$$J = \sum_{j=1}^{k} \sum_{x_i \in c_j} d\left(c_j, x_i\right)^2 \tag{3}$$

where the c_j represent the cluster center for a group of x_i, and x_i^k means the kth dimension of x_i.

However, the traditional K-means faces two main problems: the determination of the initial cluster center positions and the number of clusters K. To deal with the problem of initial center positions, a bisecting K-means algorithm can be adopted. It overcome some drawbacks of the traditional K-means algorithm and become more robust. The main difference between the bisecting K-means algorithm and the traditional K-means algorithm is that, in the former one, only one cluster center needs to be addressed in each iteration according to the sample distribution. Thus, the bisecting K-means algorithm can effectively reduce the chance of falling into local optima and the sensitivity towards noises. The detailed process of bisecting K-means is shown in Figure S12 (Supporting Information).

Though, the problem of choosing initial cluster center positions is solved, the determination of the number of clusters still remains unsettled. To deal with it, an improved bisecting K-means algorithm based on a canopy algorithm is applied. In this method, the number of clusters K is provided by the canopy algorithm, which is then used in a following bisecting K-means algorithm.

The canopy algorithm is an unsupervised preclustering algorithm. Compared to other methods such as elbow rule and exhaustive search using some evaluation metrics, canopy algorithm does not need to operate many times to search for the best cluster number K. It can generate cluster number K based on the sample distribution. Also it is much faster than other methods especially when dealing with high-dimensional large data sets. The process of the algorithm is presented in Figure S13 (Supporting Information).

In short, canopy algorithm was applied to determine the cluster number and the bisecting K-means algorithm was chosen to cluster the battery. Then the clustering result was validated with the traditional sorting method, reaching a high accuracy of 88%.

The data clustering processing including the training and testing was performed in MATLAB. While the control of the machine was performed in python using the Numpy, pandas and sklearn packages.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Author Contributions

A.R. and Z.Z. contributed equally to this work. A.H.R., Z.H.Z., X.Z., G.D.W, B.H.L., H.B.S., and F.Y.K. developed this idea and data analysis. A.H.R. and S.X.C., K.Q., P.B.N., and Z.L.L. carried out experiment and data collection. Z.H.Z. and X.Z. carried out algorithm optimization. All authors discussed the results and comments on the manuscript.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

bisecting K-means algorithms, fast clustering, lithium-ion batteries, pulse tests, retired batteries

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